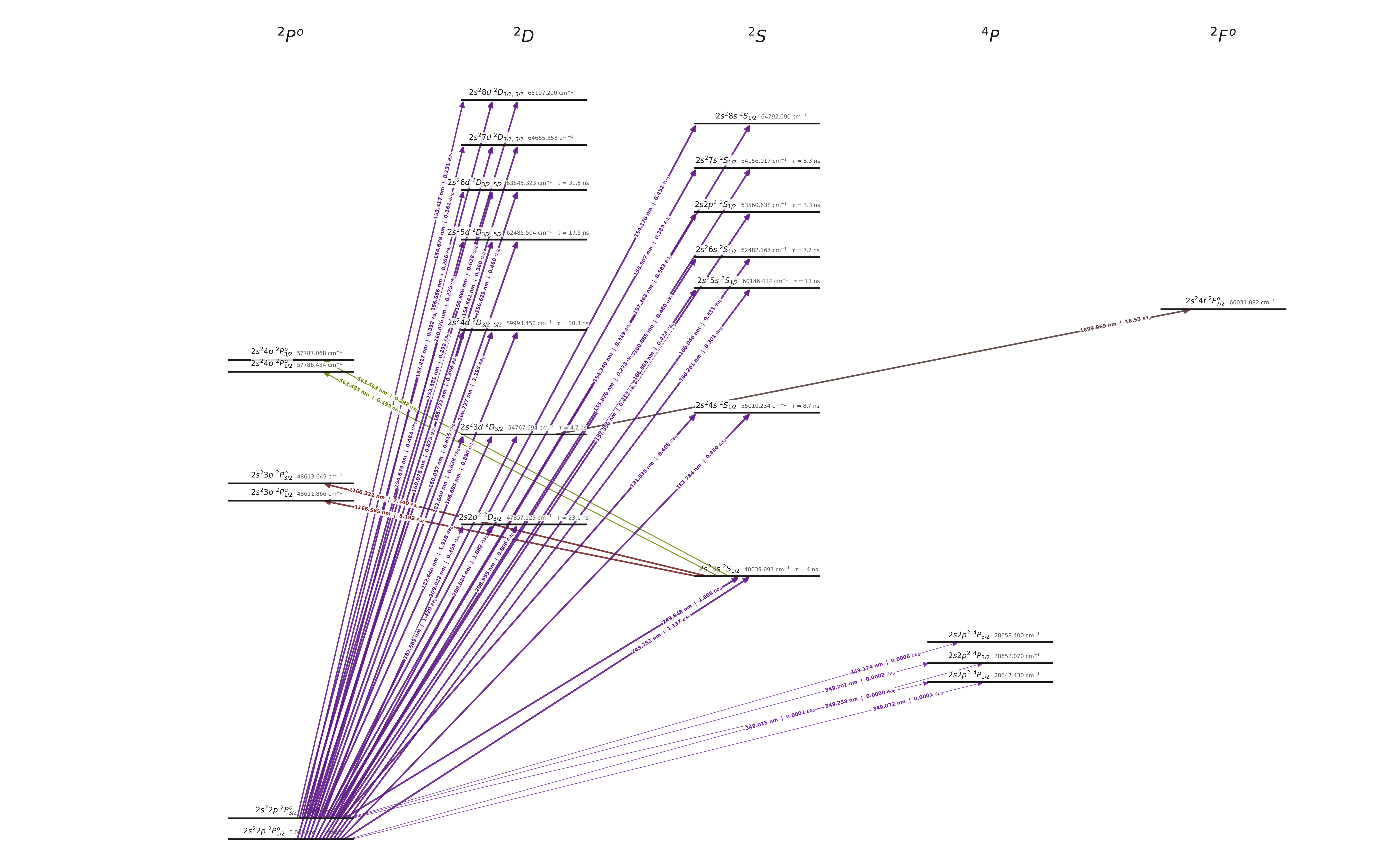


# Boron

B (B I). Strongest classified transitions below 2 μm (NIST): 31 levels, 45 lines.

B<sup>+</sup> ionisation limit 66928.04 cm<sup>-1</sup> = 8.29802 eV (149.414 nm)



## Key transitions

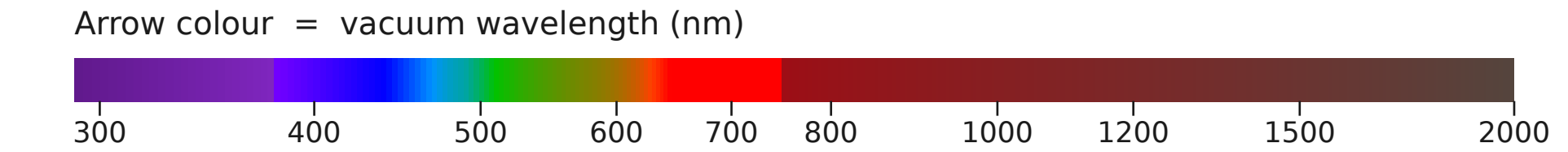
| Transition                                                                               | λ vac (nm) | λ air (nm) | ν (THz)     | ⟨J  er  J'⟩  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use       |
|------------------------------------------------------------------------------------------|------------|------------|-------------|---------------------------------|----------------------|----------------|------------|-----------|
| 2s <sup>2</sup> 2p 2p <sub>1/2</sub> <sup>o</sup> - 2s <sup>2</sup> 3s 2s <sub>1/2</sub> | 249.7523   | 249.6770   | 1200.359034 | 1.137                           | 8.400e+07            | 39.79          | 33.6       | 249.75 nm |
| 2s <sup>2</sup> 2p 2p <sub>3/2</sub> <sup>o</sup> - 2s <sup>2</sup> 3s 2s <sub>1/2</sub> | 249.8476   | 249.7723   | 1199.9014   | 1.608                           | 1.680e+08            | 39.79          | 67.2       | 249.85 nm |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## <sup>11</sup>B hyperfine structure (I = 3/2)

| Level                                             | A (MHz)      | B (MHz)    | Shift of each F level from the centre of gravity (MHz) |
|---------------------------------------------------|--------------|------------|--------------------------------------------------------|
| 2s <sup>2</sup> 2p 2p <sub>1/2</sub> <sup>o</sup> | 366.0765(15) | -          | F=1: -457.60 F=2: +274.56                              |
| 2s <sup>2</sup> 2p 2p <sub>3/2</sub> <sup>o</sup> | 73.3496(4)   | 2.6927(10) | F=0: -271.69 F=1: -201.04 F=2: -57.03 F=3: +165.71     |
| 2s <sup>2</sup> 3s 2s <sub>1/2</sub>              | 238.0(4)     | -          | F=1: -297.50 F=2: +178.50                              |

Measured A, B constants; sources in the data file.



## Isotope shifts

| Transition                                                                               | λ vac (nm) | Isotopes | Shift (MHz) |
|------------------------------------------------------------------------------------------|------------|----------|-------------|
| 2s <sup>2</sup> 2p 2p <sub>1/2</sub> <sup>o</sup> - 2s <sup>2</sup> 3s 2s <sub>1/2</sub> | 249.752    | 10-11    | 5031.3(20)  |

Shift = ν(first isotope) – ν(second isotope).